Observability's Sixth Sense: Detecting Anomalies in Metrics

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VictoriaMetrics

Why this talk?

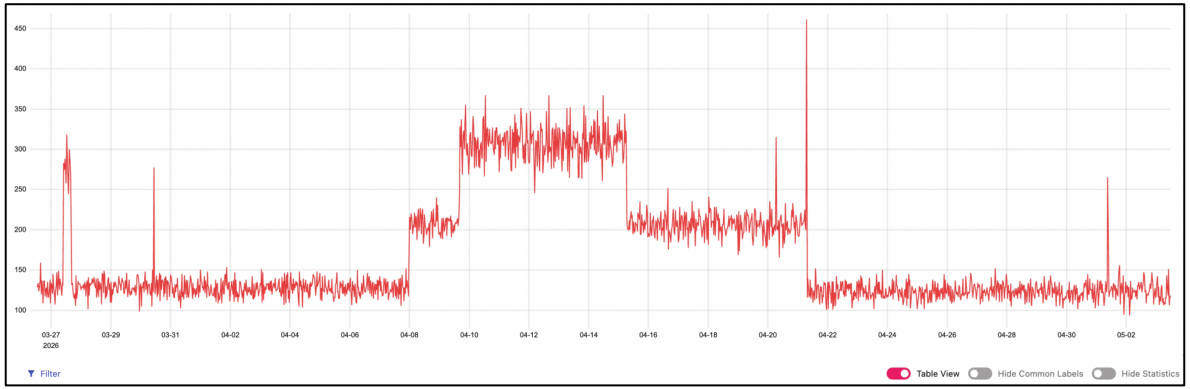
- We collect more telemetry than ever, yet debugging production issues feels reactive.
- Anomaly detection can help developers identify meaningful signals faster
- Improve MTTR
- Less time firefighting

Why should you stay and listen?

- Walkthrough through anomaly detection fundamental concepts
- Demos on using anomaly detection tools, identifying patterns, building specific dashboards

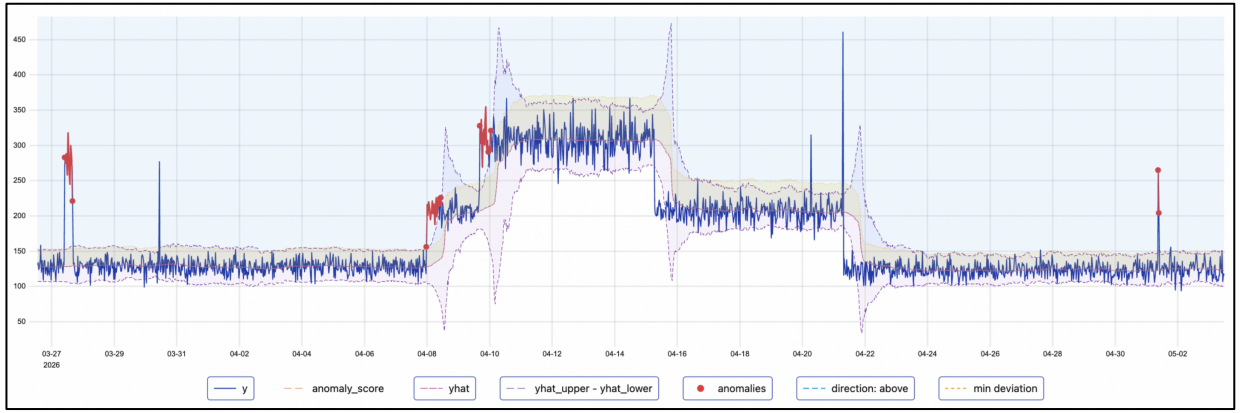
What is an anomaly?

Anomaly in **time series data** refers to a **data point or a pattern** that deviates significantly from the expected behavior.

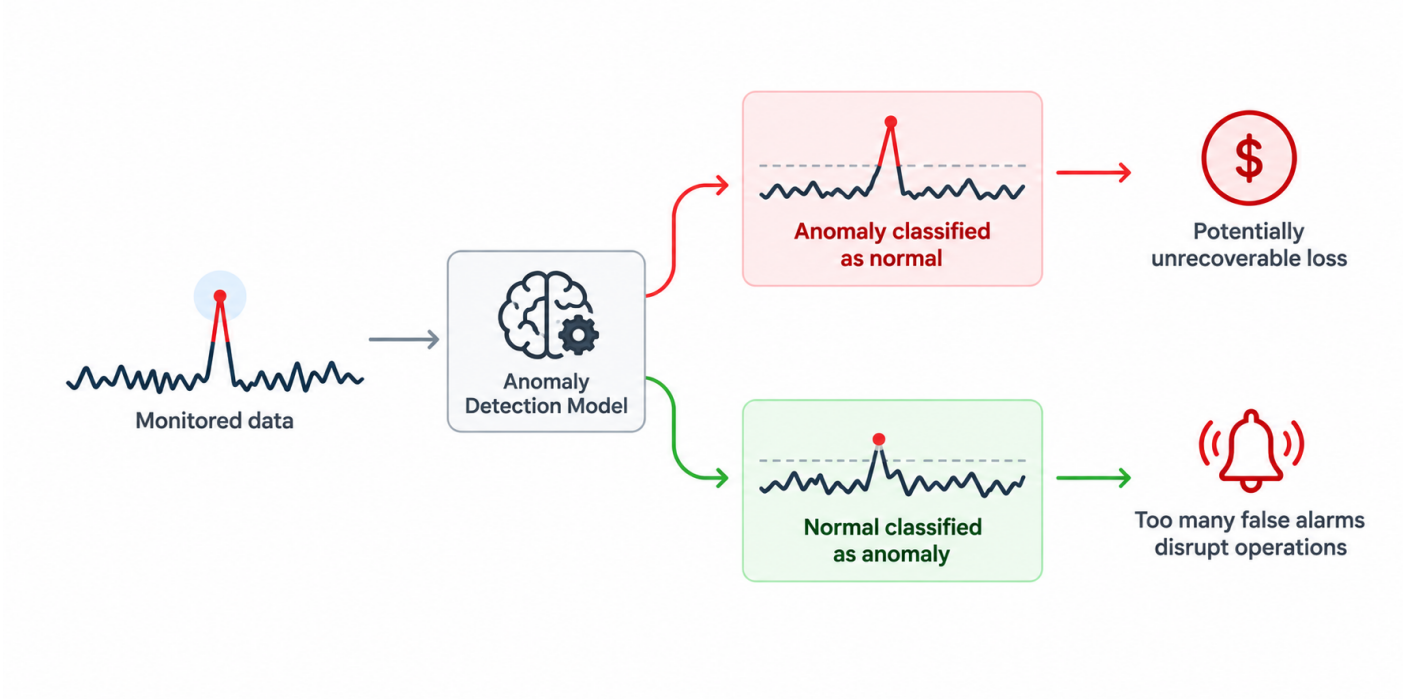


What is anomaly detection?

Anomaly detection in time series data involves the **identification and highlighting** of unusual data points, patterns, behaviors within a given time frame.



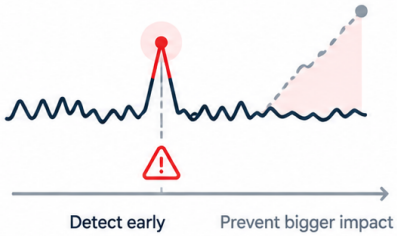
Why is anomaly detection not enough?



How anomaly detection creates business value

Detect issues faster

Identify irregularities early, before they escalate.

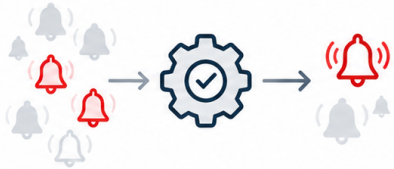


Improve operational efficiency

Find performance bottlenecks and process deviations.

Reduce monitoring overhead

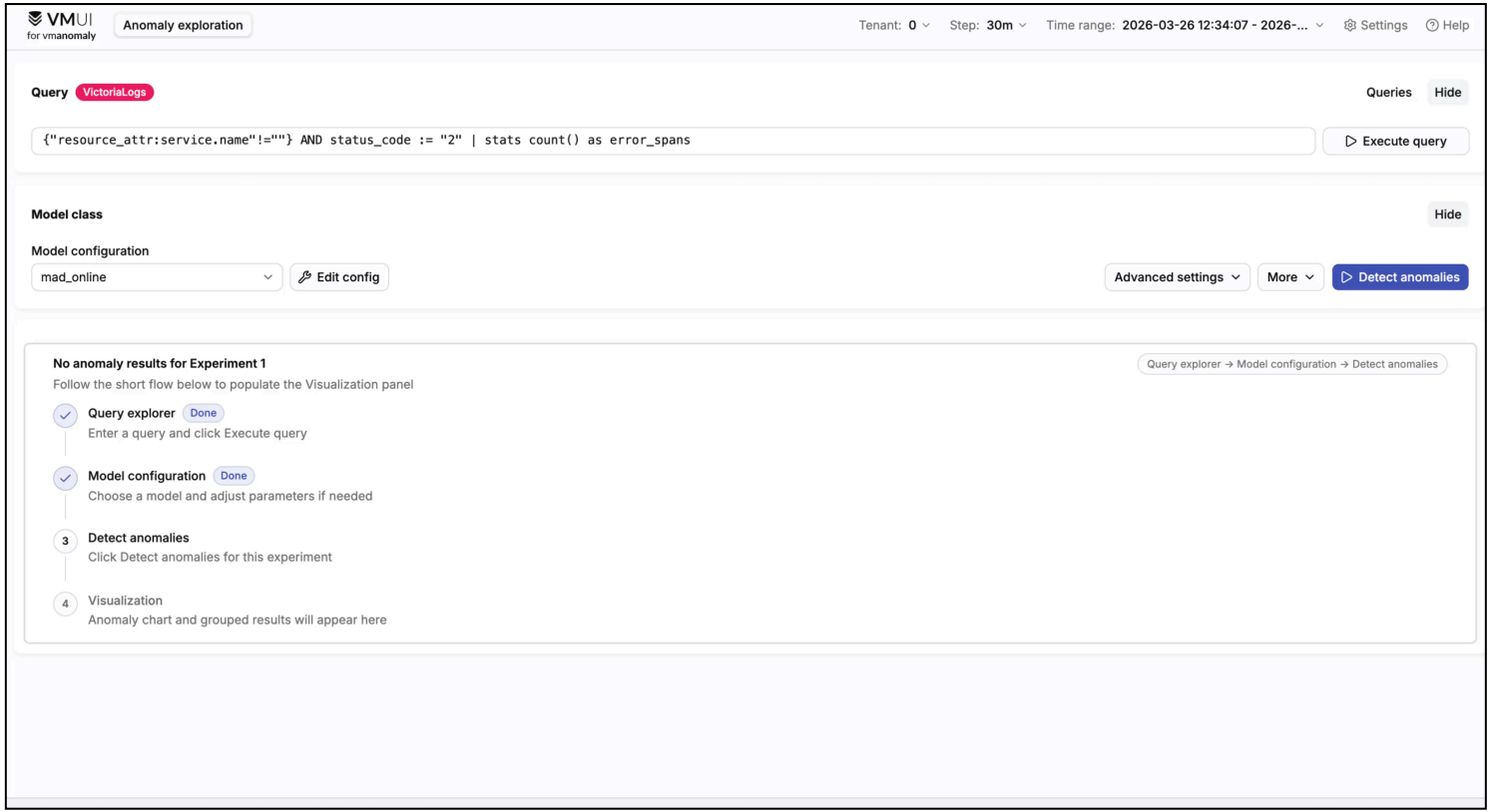
Automate rule management and filter noise.



Anomaly detection models

- Z-score: recommended for **testing**
- Rolling quantile
- MAD (Median Absolute Deviation)
- Isolation Forest: Used to address metric interaction - **multivariate** anomalies
- Prophet: Used as a **reasonable default**
- Holt-Winters
- STL decomposition
- Gradually update model behavior to new data (Z-score, MAD, SeasonalQuantile)

ARE YOU STILL THERE?

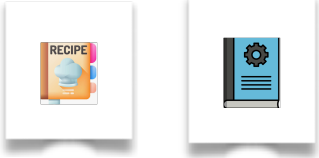

VMUI
for vmanomaly
Anomaly exploration
Tenant: 0
Step: 30m
Time range: 2026-03-26 12:34:07 - 2026-...
Settings
Help
Query
VictoriaLogs
Queries
Hide
{"resource_attr:service.name"!=""} AND status_code := "2" | stats count() as error_spans
Execute query
Model class
Hide
Model configuration
mad_online
Edit config
Advanced settings
More
Detect anomalies
No anomaly results for Experiment 1
Query explorer → Model configuration → Detect anomalies
Follow the short flow below to populate the Visualization panel
Query explorer
Done
Enter a query and click Execute query
Model configuration
Done
Choose a model and adjust parameters if needed
3
Detect anomalies
Click Detect anomalies for this experiment
4
Visualization
Anomaly chart and grouped results will appear here

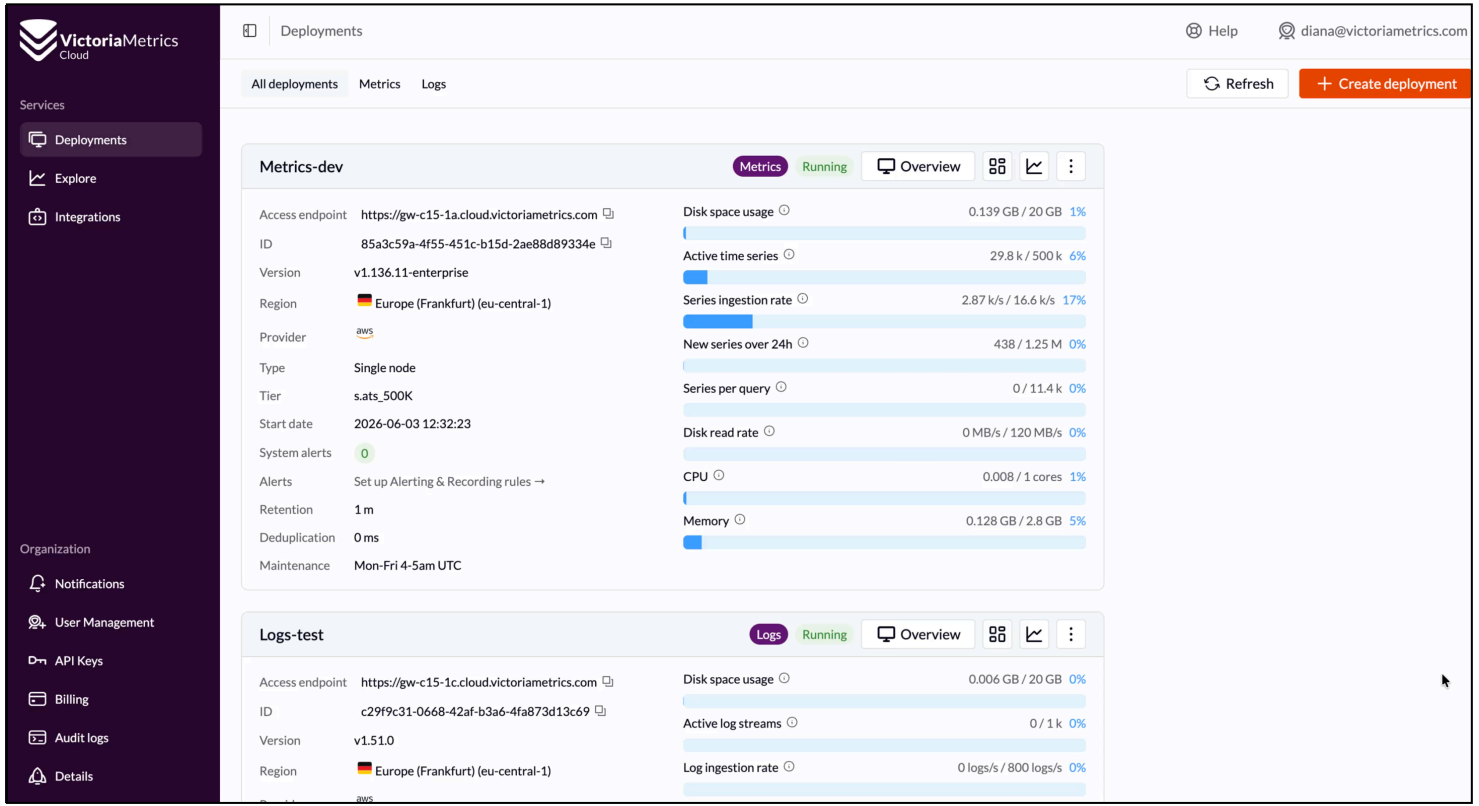
MCPs

Connects the AI agent to live tools, APIs, documentation and data. It lets the agent do things instead of merely explaining them.

Skills

Reusable playbooks that teach an AI Agent how to accomplish a particular job. They contain domain knowledge, recommended steps, guardrails and examples.




VictoriaMetrics Cloud
Deployments
Help
diana@victoriametrics.com
Services
Deployments
Explore
Integrations
All deployments
Metrics
Logs
Refresh
Create deployment
Metrics-dev
Metrics
Running
Overview
Access endpoint https://gw-c15-1a.cloud.victoriametrics.com
ID 85a3c59a-4f55-451c-b15d-2ae88d89334e
Version v1.136.11-enterprise
Region Europe (Frankfurt) (eu-central-1)
Provider aws
Type Single node
Tier s.ats_500K
Start date 2026-06-03 12:32:23
System alerts 0
Alerts Set up Alerting & Recording rules →
Retention 1 m
Deduplication 0 ms
Maintenance Mon-Fri 4-5am UTC
Disk space usage 0.139 GB / 20 GB 1%
Active time series 29.8 k / 500 k 6%
Series ingestion rate 2.87 k/s / 16.6 k/s 17%
New series over 24h 438 / 1.25 M 0%
Series per query 0 / 11.4 k 0%
Disk read rate 0 MB/s / 120 MB/s 0%
CPU 0.008 / 1 cores 1%
Memory 0.128 GB / 2.8 GB 5%
Logs-test
Logs
Running
Overview
Access endpoint https://gw-c15-1c.cloud.victoriametrics.com
ID c29f9c31-0668-42af-b3a6-4fa873d13c69
Version v1.51.0
Region Europe (Frankfurt) (eu-central-1)
Disk space usage 0.006 GB / 20 GB 0%
Active log streams 0 / 1 k 0%
Log ingestion rate 0 logs/s / 800 logs/s 0%
Organization
Notifications
User Management
API Keys
Billing
Audit logs
Details

I have a cloud deployment with ID 85a3c59a-4f55-451c-b15d-2ae88d89334e, I need a metrics overview.

Thinking about retrieving and summarizing cloud deployment metrics data

Now let me pull key resource and ingestion metrics for the deployment.

Used victoriametrics integration

All 7 scrape targets are up, so I have a good health picture. Here's the overview:

Deployment Metrics Overview — `85a3c59a-4f55-451c-b15d-2ae88d89334e`

Scale & storage

- Stored rows: **~74.2M**
- On-disk data size: **~38.0 MB**
- Free disk space: **~13.5 GB**

Ingestion

- Insert rate: **~161 rows/sec** (5m avg)

Resource usage

- Resident memory: **~790 MB**

Health

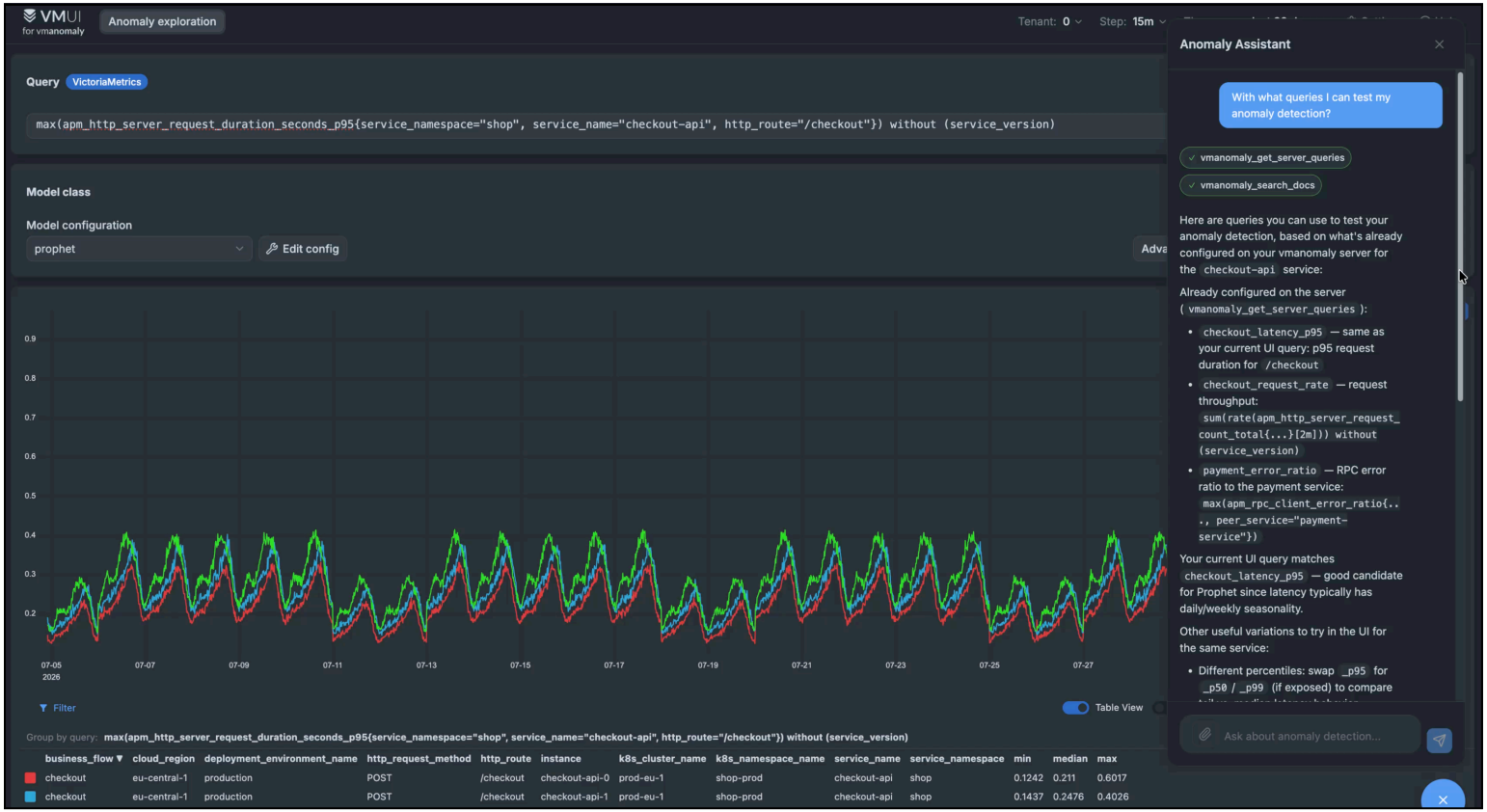
- Scrape targets: **7/7 up** (no targets down)

Metric catalog

- Total distinct metric names available: **~750+**, dominated by Kubernetes/cluster metrics (`apiserver_*` , `kubelet_*` , `container_*` , `coredns_*`) plus native VictoriaMetrics internals (`vm_*` , `vmagent_*` , `vminsert_*` , `vmselect_*`) — this looks like a deployment scraping a full k8s cluster in addition to self-monitoring.
- Query usage stats show a sample of tracked metrics with **0 query requests** recorded in the current stats window (since roughly early Feb 2026, based on the collection start).

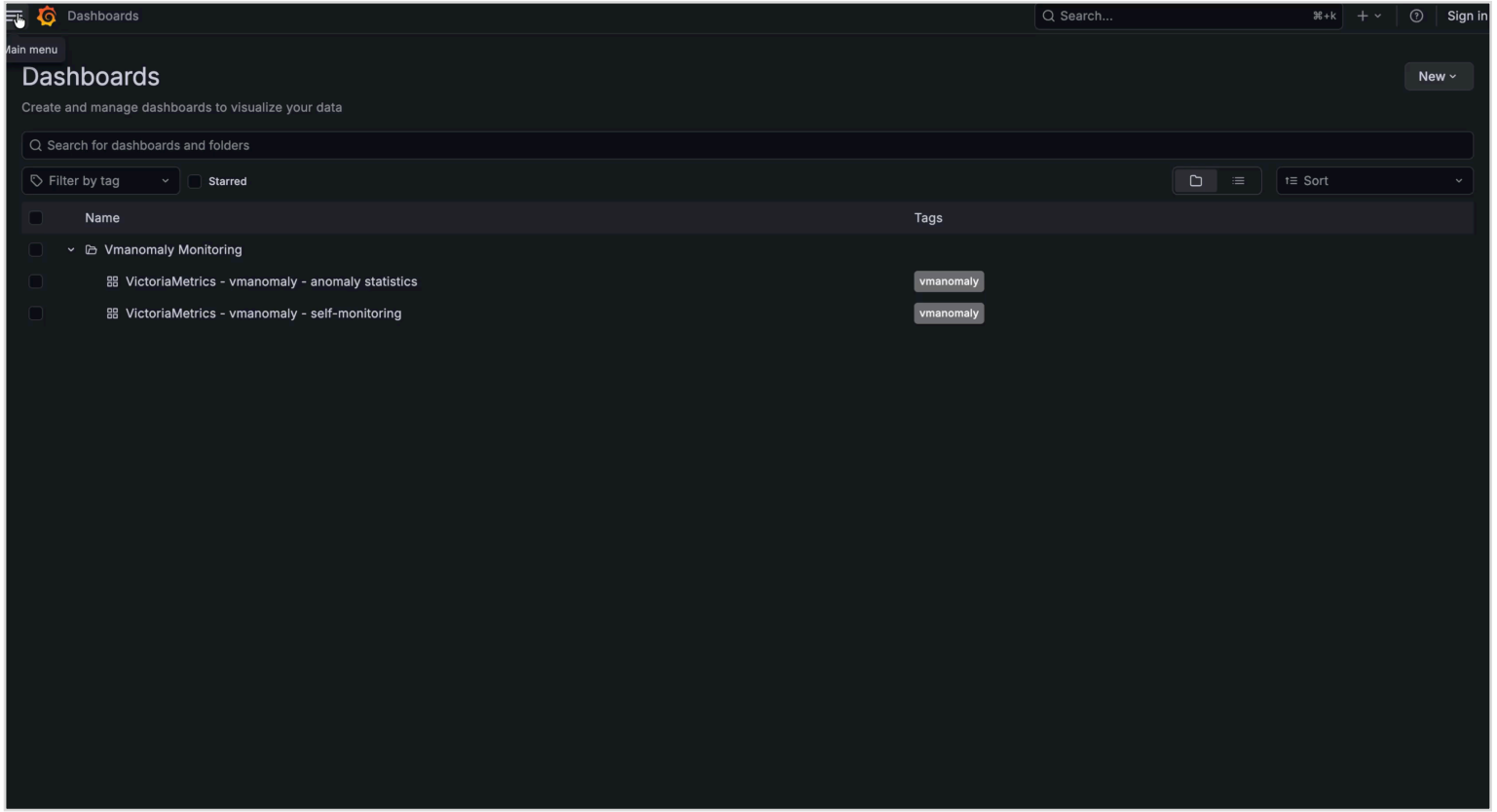
Write a message...

Sonnet 5 Medium


VMUI
for vmanomaly
Anomaly exploration
Tenant: 0
Step: 15m
Query
VictoriaMetrics
max(apm_http_server_request_duration_seconds_p95{service_namespace="shop", service_name="checkout-api", http_route="/checkout"}) without (service_version)
Model class
Model configuration
prophet
Edit config
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
07-05
2026
07-07
07-09
07-11
07-13
07-15
07-17
07-19
07-21
07-23
07-25
07-27
Filter
Table View
Group by query: max(apm_http_server_request_duration_seconds_p95{service_namespace="shop", service_name="checkout-api", http_route="/checkout"}) without (service_version)
business_flow
cloud_region
deployment_environment_name
http_request_method
http_route
instance
k8s_cluster_name
k8s_namespace_name
service_name
service_namespace
min
median
max
checkout eu-central-1 production POST /checkout checkout-api-0 prod-eu-1 shop-prod checkout-api shop 0.1242 0.211 0.6017
checkout eu-central-1 production POST /checkout checkout-api-1 prod-eu-1 shop-prod checkout-api shop 0.1437 0.2476 0.4026
Anomaly Assistant
With what queries I can test my anomaly detection?
vmanomaly_get_server_queries
vmanomaly_search_docs
Here are queries you can use to test your anomaly detection, based on what's already configured on your vmanomaly server for the checkout-api service:
Already configured on the server (vmanomaly_get_server_queries):
checkout_latency_p95 — same as your current UI query: p95 request duration for /checkout
checkout_request_rate — request throughput: sum(rate(apm_http_server_request_count_total{...}[2m])) without (service_version)
payment_error_ratio — RPC error ratio to the payment service: max(apm_rpc_client_error_ratio{..., peer_service="payment-service"})
Your current UI query matches checkout_latency_p95 — good candidate for Prophet since latency typically has daily/weekly seasonality.
Other useful variations to try in the UI for the same service:
Different percentiles: swap _p95 for _p50 / _p99 (if exposed) to compare
Ask about anomaly detection...

Monitor your anomaly detection backend

Github repository


Dashboards
Search...
⌘+k
Sign in
Main menu
Dashboards
New
Create and manage dashboards to visualize your data
Search for dashboards and folders
Filter by tag
Starred
Sort
Name
Tags
Vmanomaly Monitoring
VictoriaMetrics - vmanomaly - anomaly statistics
vmanomaly
VictoriaMetrics - vmanomaly - self-monitoring
vmanomaly

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Take-aways

- Anomalies in time series appear as spikes, drops, drift, or missing patterns—not all are meaningful
- Interpreting anomalies requires system and business context, not just metric deviation
- Anomaly detection is useful for surfacing unknown or gradual issues, but misleading when applied blindly
- Effective use depends on selective application and context-aware practices, alongside dashboards and alerts

Teşekkürler

Join our community!

https://play-vmanomaly.victoriametrics.com/metrics/vmui
https://play-vmanomaly.victoriametrics.com/traces/vmui
https://docs.victoriametrics.com/anomaly-detection/
https://victoriametrics.com/blog/observabilitys-sixth-sense-grounding-anomaly-detection-in-reality/
https://github.com/VictoriaMetrics/mcp-vmanomaly
https://github.com/jgomezselles/cnd-workshop-2026/tree/main/anomaly-detection